7. (a) Find the mass of the tetrahedron plate bounded by the coordinate planes and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$, the variable density $P=\lambda xyz$.

(b) Express $\int_0^1 x^m\left(1-x^n\right)^p dx$ in terms of Gamma function and hence evaluate $\int_0^1 x^5\left(1-x^3\right)^{10} dx$.

8. (a) Obtain the Fourier series for $f(x)=e^{-x}$ in the interval $0<x<2\pi$.

(b) Find the Fourier sine and cosine series for $f(x)=x$ in the interval $[0,\pi]$. Hence show that $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots=\frac{\pi^2}{8}$.

$\cos(\frac{\pi}{2}+\pi)$
$=-\cos\frac{\pi}{2}$

[BENG – 1129]

I/IV B.Tech. DEGREE EXAMINATION.

First Semester

MATHEMATICS – I

(Common for Group-A and Group-B branches)

(Effective from the admitted batch of 2022–2023)

Time : Three hours Maximum : 70 marks

Answer question No.1 compulsorily and any FOUR questions from remaining.

All questions carry equal marks.

All parts of a question must be answered at one place only.

1. (a) If $u=x^4y^5$, $x=t^2$, $y=t^3$ then find $\frac{du}{dt}$.

(b) State Rolle's mean value theorem.

(c) What do you mean by saddle point?

(d) Find $\Gamma(-3.5)$.

(e) Change the integral $\int_{\partial}^{a}\int_{y}^{a} \frac{x}{x^2+y^2}\,dx\,dy$ into polar coordinates.

(f) Write the conditions for the expansion of a function as a Fourier series.

(g) Is the function $f(x)=\begin{cases}1-x, & -\pi < x < 0\\ 1+x, & 0 < x < \pi\end{cases}$ is even or odd? Justify.

2. (a) If $z=\frac{x^2+y^2}{x+y}$ then show that $\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2 = 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)$.

(b) If $f(x,y)=\frac{1}{x^2}+\frac{1}{xy}+\frac{\log x-\log y}{x^2+y^2}$ then prove that $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}+2f=0$.

3. (a) If $u=f(y-z, z-x, x-y)$ then prove that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=0$.

(b) Expand $F(x,y)=x^2y+3y-2$ is powers of $(x-1)$ and $(y+2)$ by Taylor's theorem.

4. (a) Determine the points where a function x^3+y^3-3axy has a maximum or minimum.

(b) Find the maximum and minimum distances of the point $(3,4,12)$ from the sphere $x^2+y^2+z^2=1$.

5. (a) Change the order of integration is $\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}\,dy\,dx$ and hence evaluate.

(b) Evaluate $\iint r^3\,dr\,d\theta$ over the area between the circles $r=2\cos\theta$ and $r=4\cos\theta$.

6. (a) Find the volume bounded by the xy-plane, the paraboloid $2z=x^2+y^2$ and the cylinder $x^2+y^2=4$.

(b) Find by triple integration the volume of the solid bounded by the surface $x=0, y=0, x+y+z=1$ and $z=0$.